

be using digital twin technology in production.

Digital twin technology is valued presently at US\$ 1.88 billion and is expected to have an exponential growth to reach US\$ 13.5 billion by the end of 2025, growing at a CAGR of 32.4 per cent during the forecasted period. This smart technology is one of the most profitable technologies at present. However, rising cyberattacks and less awareness about it may affect the growth of digital twin technology in the coming years.

Recently, the US government allocated a large portion of their budget to digital twin technology for implementation in defence sector (especially in modular advanced armed robotics, XM2010 sniper weapon system, etc).

In 2015, companies such as Xerox and HP brought heavy-duty printers into the IoT environment to facilitate their maintenance. To take this further, digital twin can improve design of such products to a great extent. Digital twins use data, machine learning and the IoT to help companies optimise, innovate and deliver new services. Certainly digital twin is the future of Industry 4.0.

Advantages

Industry can derive major advantages by using digital twin technology, some of which are mentioned below.

- Monitor a constant stream of usage and performance data in real time
- Combine end-to-end asset or product life cycle data into digital threads
- Support new products as a service business model
- Drive innovations in manufacturing, R&D, supply-chain management, service and logistics

Digital twin in healthcare sector

With digital twin, lives can be improved in terms of health, sports and education by taking a more data-driven approach to healthcare. Healthcare is rapidly embracing digital twin technology. The main aim here is to deliver data-driven personalised medicine.

Digital twins are built on computer-based or crypto models that can create individual and group data for medical purposes. Researchers are aided by these digital representations of human physiology in their studies of diseases, new drugs and devices. Digital twins also help to accelerate medical innovation.

In future, these tools may even help doctors accurately optimise the performance of patient-specific treatment plans. In short, this technology can bring life-saving innovation to market faster, reduce medical cost and increase patient safety.

Digital twins are helping in research of tumours and development of new drugs. A digital twin virtualises a hospital system to create a safe environment for testing the impact of a potential change on system performance. Healthcare delivery is massively complex. Today, common-sense, spreadsheets and statistics are just not enough for strategic decisions to be taken by doctors. By creating a digital twin of a hospital, they can observe potential changes in operational strategy, capacities, staffing and care delivery models. **EFY**

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